ModelIt!

Student Activity Pack

What Makes a Shadow?

The Science of Light and Shadows

Grade 1 | NGSS 1-PS4-3 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw yourself standing outside in the sun with your shadow. Where is the shadow?

2 Circle what you need to make a shadow: light, an object that blocks light, darkness behind the object, a sandwich.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Light Position | Shadow Size

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Light Close to Object

Light Position set to close to the object

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Light Far from Object

Light Position set to far from the object

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What makes a shadow appear, and why do shadows change shape? Use evidence from your simulation to support your answers.

8 STEM Challenge: Shadow Puppet Show

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YOUR ENGINEERING MISSION

Design shadow puppets and use a flashlight to put on a short shadow puppet show for the class.

THE SCENARIO

Your class is hosting a shadow puppet theater! Cut out fun shapes from cardboard, attach them to sticks, and figure out how to make big shadows and small shadows on the wall. Use what you learned about light and distance!

Engineering Design Thinking

1. WHAT SHAPES WILL YOU CUT OUT FOR YOUR SHADOW PUPPETS?
2. HOW WILL YOU HOLD YOUR PUPPETS BETWEEN THE LIGHT AND THE WALL?
3. HOW CLOSE SHOULD THE LIGHT BE TO MAKE THE BIGGEST SHADOW?
4. CAN YOU MAKE YOUR PUPPET'S SHADOW MOVE AND TELL A STORY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Lighting Designers create the perfect lighting for movies, theater shows, and concerts. They use lights and shadows to make scenes look amazing. They earn $40,000-$85,000/year.